

You may be more excited than I am for today as a midterm test My excitement starts afterwards when I'm grading your work But a bit of excitement still because I hope all logistics works out as I plan them what we'll see So please sit in the room assigned to you There's plenty of time.

I will not run over time today because actually should be a bit tiny bit shorter than two times 45 minutes But today we're going to start on module 6 on light the nature and propagation of light Before I do that.

However, let's briefly recap a few remember how waves work waves on a string standing waves on a string So this was I think an exam question two years back final test order reason I can't remember This is a type of question you should be able to do and this indeed may involve a bit of calculation But you should be able to do this Sometimes it helps to make a drawing Have a string fix at two ends.

So it can have this this mode can have this mode Can have this mode Etc right and being a number of anti nodes And there's one and it's two and it's three etc.

Are we ready to vote? Not yet Let me try to guide you through answering this of this expression Which of these parts denotes the spatial behavior which denotes the time behavior second part denotes $\sin(\omega t)$ so that's a temporal time behavior So the spatial part is the only thing you need to answer this question because this question is about the spatial behavior of this string That already rules out all the answers with an ω in it because that relates to the time behavior, right? Okay, so then there's four left Because all of these modes will vibrate in time and they have a sign is all the time behavior. So that does not Dictate how the string behaves in in a spatial sense, right? So of the four answers we still have left the ones without ω .

So that's C D E and G Who would vote for C D E Few others and what was lost on G and there's still No consensus yet Okay.

How do we solve this and $\sin(kx)$.

That's the spatial behavior. I need fixed points at x is zero.

I need $\sin(kx) = 0$ that's satisfied by default, right? But also at $x = L$.

I need $\sin(kL) = 0$ To be zero This happens if this argument is multiples of π So this happens if $kL = n\pi$ where n goes from 1 2 3 etc And in this way I can express L as $n\pi/k$ So which which of these is $n\pi/k$? So answer G would be the correct one Okay So This is assumed prior knowledge that you know when the sign is zero. It's calculus, right? But this is how to solve this question, okay Question yeah, so that's yeah, too, but you have if you can express it in that way in terms of Wave number for instance, that's also fine.

This is the spatial behavior No, it should this is the only correct answer there should not be an ω because ω Does not that describes only this Let me rephrase.

Let me start again This the a standard wave is a superposition of a forward traveling wave and a backward traveling wave at exactly the right Frequency as exactly the right velocity and after combining that you end up with a Description that exactly looks like this for two fixed ends. You have the fundamental modes and the higher modes, etc So sign as a function of x This could be a cosine if you want But if you'd start from rest it starts like this and this dictates the spatial behavior. So it's You may use the wave velocity as an intermediate step But ultimately if you write it back to the given parameters you end up with answer G

Okay Let me continue. So That's what we got last time wave equation which was a scary-looking thing here of partial derivatives and There's solutions to this forward traveling and backward traveling waves Look at that overlapping waves and standing waves So this is what well the key takeaways were of last the last lecture today We're going to look at lights and how that propagates and what I would behaves So a lot of this will be repetitions snails laws you've seen in secondary school, but there's also some nice discussion So already before Christ People knew that lights could reflect from smooth surfaces and the philosophers in the with the ancient Greeks Looked thought that lights while we saw things by sending out rays From our eyes which then reflected on your on the object and in that way we could well detect everything around us also the ancient Greeks observed that Light can be refracted, which is bending at an interface between two boundaries That have different material properties different refractive indices So this started already well a long time ago Then in the 17th century, there was a debate about the nature of light We had Newton well the celebrated physicist who well basically developed module two for you And we had Christiane Huygens a Dutch guy And they were in conflict one thought light was a particle the other one thought it was a wave And well the truth is it's actually both and it chooses Well, you can look at different interpretations to explain different phenomena I'll get back to that on the module seven on quantum mechanics people try to measure the speed of light and they do did this using astronomy So by looking at an eclipse of moons of in this case Jupiter So the moon IO of Jupiter they made an estimate that light was two point one ten times ten to the eight meters per second Which is a pretty decent estimate. I would say for that time Almost 200 years later They refined that to a better estimate. This was only well five percent off and then Maxwell Predicted electromagnetic waves that would be traveling at the speed of light So that's a known constant speed of light the vacuum is almost three times ten to the eight meters per second and then Heinrich Hertz German guy Demonstrated that this did not only hold for light but also for any electromagnetic wave So not just a visible spectrum, but any electromagnetic wave now we had Einstein Who started the discussion on whether light was a particle or a wave and he quantized light? In terms of photons light particles he defined them and he needed that to explain the photoelectric effects which I'll also come back to in detail in quantum physics and then after quantum mechanics they Well, let's say agreed on that light light is both both a particle and a wave and depending on what phenomena Phenomenon you are explaining you need either one interpretation or the other but never do at the same time So absorption and emission of light that's best described by particles because this feels like a discrete thing a light Particle is absorbed and then reemitted Whereas propagation and also interference for that you really need the wave behavior of light And I'll come back to interference and diffraction for instance in the lecture on Monday next week Well, of course XKHD has something to say about light interpretation interpreting physics there's two viewpoints and probably The average is the truth. Well that holds for this example holds also for light but this is Well could be thought of as coincidence, but it's a it is the case Well to get more to the real content of today's lecture There is a whole spectrum of electromagnetic waves. Of course, there's the very low frequencies with wavelength in the order of meters and you have AM radio all the way up to Well Cell phone communication, which is Well in the gigahertz range.

There's the terahertz range where there's visible light and Then you can go to ionizing radiation with a radiation that really damages cells because it can Get electrons loose from the atoms and then you are talking about ionizing radiation, which is really harmful to the human body Another way to look at this is here.

So there's a radio spectrum. There's a microwave spectrum terahertz And then we get to light infrared ultraviolet in between. There's the visible light, of course And then you go to the ionizing radiations x-rays soft and hard x-ray, which is really damaging to tissue and then gamma radiation So this probably you know Propagation of lights holds for all of this But we typically confine ourselves to this narrow

region because we talk about visible light. Also, when we talk about rainbow formation, it's well, we look at the colors in visible light. Of course, well this range goes from well, what is that three hundred nanometers to? well seven hundred nanometers. And that's what we call visible light.

There's infrared here, ultraviolet here, and the eye has three cones in the retina that are sensitive to different colors of the light. And these well kind of overlap.

It's designed by nature in this way. So we have to work with this. So we can distinguish between blue, green, and red colors, and their combination gives us all the visible spectrum. So so much for the introduction. Let's go to how to work with this. So to describe how light propagates, we need the concept of wavefront and light rays. So a wavefront. Consider like a point source in space that radiates light in all directions, and then the wavefront is like well, if it sends out a wave, you can think of where the wave is. The peak of this wave is how that is propagating in space, and you could consider that a wave front that propagates outwards in radial directions. What characterizes these points is that they all have the same phase. So it's always a peak of a wave that is propagating outwards. So if we if I draw the locus, so the collection of all the points that have that same phase, I would get a spherical shell for a point source. But this holds in general. I can draw a wave front as being the locus of all adjacent points at which the phase of vibration is the same. We talk about light, this could also be waves on the string, waves on the pond. Then the wavefront is this. If you throw a rock in the pond, then you see the circle spread, and then the first circle spreading that is a wavefront spreading in a circular way, right?

And this could be a sound point source, but can be any source basically that generates waves, and for any way if you can define wavefronts, right? Next to that, next two wavefronts.

We have rays. A ray is a line that I'll say a beam of light propagating in a certain direction. So at every point on a wavefront, a light ray indicates in what direction this wavefront is moving, and if we look at homogeneous isotropic media, which is what we will confine ourselves to in this course. Then we have the rays always perpendicular to the wavefronts. If a wave is propagating outwards, say that the circular example, you throw a rock on a pond, and it spreads cylindrically, then the rays in which it spreads are from this point outwards in radial direction, and the wave fronts are circles, and then by definition.

Well, so that's kind of a top-down view of what we see here. The rays are always locally perpendicular to the wavefronts. And if you look at the description of light in terms of rays, and you talk about the field of geometrical optics, where you just mainly look at geometry and how that influences light rays. That's basically all of the calculations. You've done at your secondary school. With light going through prisms, through blocks of glass, for instance, all of that is based on geometrical optics. We will also confine ourselves to that. Because those problems are difficult enough already. They use a lot of trigonometry, which is often a challenge if you have a source that's really far away. Then well, you can. The source has a spherical spreading of the light rays coming from that source. So the wavefronts are spheres, but if you're far away, then locally, where you're looking at these rays, arrive kind of? Well, just as plane waves, right? If you're far enough away, then consider this a zoomed-in enough part of this sphere, where locally you just have parallel wave fronts, and then that well, then you have a plane wave, which is the approximation we often make. Also if you have a very focused light source, like a laser. It emits a small, confined beam, and within that beam well, there's wave fronts that are perpendicular to that beam. Okay. And this is called the plane waves, that's the assumption we typically make. We look at spreading of propagation of rays. For interference and diffraction, it's different. We need spherical sources or

cylindrical sources But in general what we do geometrical optics We have wave fronts perpendicular to the rays and these are all parallel. So we consider plane waves. Okay can we Have two wave fronts from the same wave cross one another who thinks yes who thinks no The rest doesn't have an opinion yet Indeed it is no Because each wave front represents a certain phase So if you cannot have if you throw one stone in the water, you cannot have two wave fronts With the same phase cross each other.

There's a unique set of Wave fronts coming out from that from that source They can never cross because at the point where across it would have two phases, which is in non unique description So you can't have wave fronts Crossing themselves, right Double slit experiment.

No, then you have two wave fronts two sources and then you can do interference Yeah, so after the double slit experiments, which I'll cover on next Monday and also on Thursday next week There's two sources and then you have two sources that generate wave fronts and then you can make an interference pattern They don't have the same phase No, no, it has propagated to the mirror and has been reflected is coming back soon Yeah, so every phase changes every two π multiples, right? so it propagates a propagate back and then it has To wavelength difference for instance, and then it doesn't have the same phase The same absolute phase relative.

Yes There was a question there.

Yeah Yeah, I'll get back to that in detail as well in this lecture Yeah, but also that is a way to to to construct wave fronts You make the locus of all of these Huygens sources to generate the the new wave front basically But it's still the same source. It's it's still the assumption that there's one plane wave incident on the medium that you're looking at Let me get to well the important part reflection and refraction So as a general example, I think this is a picture from the book the previous version of the book this woman outside Losing her head in due to wind and then she sees her head reflected in the window And also these guys on the inside drinking coffee see through the window this Hat outside so light goes through glass, but it's also partly reflected by glass And we need to describe Well how that works and how to do calculation on that so we need Reflection and refracted refraction so reflection and transmission to describe what is happening here Have you ever thought of why a mirror seems to flip only left right and not up down? If you look in the mirror, you just see that right mirror, but never up down. Why is it? Anyone who has a clue you can change the angle whatever you want. You will never see yourself flipped upside down No upside down so that yeah, if you do like this then Still that right is flipped right so how does it work? No, that's not the reason if it's concave yeah, then you get distortion of the image, but still left right Yeah, then you talk about lenses. I'm talking about flat mirrors No She closed one eye you still see left right flip Yeah, there's no wrong answers here. This is really a tricky question Kind of on track yeah, but that's that feels like flipping left right What no the mirror will not do that for you Yeah Yeah, that is a difficult way of explaining the real answer, but you're on the right track actually you flip forward backwards in the mirror and Due to how you perceive that you see left and right flip it's front and back of flips You can never see your back back of your head in the in the mirror unless you have a second mirror but it flips front and back and we perceive that as Flipping left and right because you look at the mirror front and then it reflects back at you front back, so that's Well isn't an eye-opener literally For how mirrors work there's more questions like this later Yeah, I hope this was clear, so it does not flip up down left right it flips front and back and that makes us perceive it as flipping left right But if you hold an object facing forward you will never see the back of that object You will see the forward side of that object reflected in the mirror Okay, let's look at the schematic view if we have typically we have Reflection and refraction so we have two media that we need typically called medium one medium B Sorry medium a

medium B There's an incident ray Part of that gets refracted so bent and transmitted and part of it gets reflected We can do this for many rays But we assume that we have a plane wave with a wave front like this incident on the interface and then it propagates also with plane Wave fronts like that and it's reflected also with plane wave fronts in that direction So just for convenience we often draw only one ray and that's how the next pictures will look like I just draw one ray that gets refracted and reflected So and we also need angles to describe this and If there's an interface between two surfaces, there's always a normal to the interface So a line perpendicular to the surface even for curved surfaces locally at every point on a smoothly curved surface There is a normal a unique definition of a normal pointing outward from that surface perpendicular to the surface locally And then angles are always measured with respect to this normal so there's the incident angle θ_i and the reflected angle θ_r and If there's refraction, there's also the θ_t And this is what you know already from Snell's laws On a in general, I think the angle of incidence is the angle of reflection Always measured relative to the normal to the surface, right? Yeah, sometimes Either the book or I give the angles not with respect to the normal But you have to bear in mind that you have to calculate the angle with respect to the normal to do Well angle of incidence equals angle of reflection, right? That depends on friction and surface properties of the ball, but if you angle the ball at the wall, then it will Yeah, we don't use light for that You would need the law of conservation of momentum to do calculations on that There was a 2d problem on that with billiard balls in the instructions, I think so you with Such calculations you would find that for probably for completely elastic collisions the angle of incidence is the angle of reflection But in general, that's not the case For light it's it always holds How this happens depends on the smoothness of the surface if you have a very smooth surface You get so-called specular reflection.

Is it that the case we will consider? well always basically where the angle of incidence is the angle of Reflection so the surface is locally smooth enough to allow for this But if you look microscopically, it can be very diffuse. It can be well, if you Go to high enough frequencies and too small enough a subject or an object that you're shine light on you Will see individual atoms and molecules that affect the reflection We typically look at this case only so we have smooth surfaces Such that we we have This that holds the angle of incidence is the angle of reflection Okay Yeah, so we mainly work with specular reflection.

So please try to summarize this for yourselves these concepts I think they are known already. I'm not sure if you used exactly this terminology Would be good to write this down for yourselves And then we'll start doing calculation on this I'm just out of curiosity curiosity for whom was light or early covered in secondary school Okay for most of you That's good, we'll do a recap anyway some examples of specular Reflection versus diffuse reflection can machine here if there's a pond in the mountains. There's no wind You really get a nice mirror image And in this case well There's asphalt.

There's rain. It's very diffuse. So you get a lot of rays Reaching your eye from different paths Not just a single path. So here is just this part. It's reflected exactly here and then you see it.

So let's Set up how to do calculations with this For that we need a property of the material. It's called the index of refraction so if you have two media There's always an interface between them So medium a medium B There is an interface between them And then if there's an interface and we assume locally we can we can We have a flat Interface and then locally we can define in normal vector That's perpendicular to the interface Every material has optical properties, which is so called Which is a so-called index of refraction denoted by n Sometimes also called refractive index, which is a property of a medium that specifies how Fast light propagates

through that medium and this is relative to the speed of light in vacuum So there's nothing faster than speed of light in vacuum. So that's c And the index of refraction is c over the velocity of light in a certain medium So if the index of refraction is 2 Then the light in the medium is goes twice half as fast Sorry goes half as fast as the speed of light in vacuum So it slows down by 50 percent.

I'll come back in detail while light slows down in a second So there's the index of refraction and typically it ranges from one upwards depending on the material you're looking at so speed of light and vacuum is well approximately three times ten to the eight meters per second and If the index of refraction is defined as follows then we can Well think of what happens in a medium the speed of light Changes, but the frequency does not change. So that is a property of light So if you shine blue light through glass, it stays blue light So the frequency does not change because that this well gives the color of light But the wavelength then has to change because the velocity equals the wavelength times frequency So if v changes and frequency stays constant and λ also has to change Right λ inside the medium is also scaled by the refractive index So the wavelength is smaller in the medium that has an n higher than one So if light enters glass from vacuum to glass You decrease the velocity you decrease the wavelength the frequency does not change just the wavelength and the velocity Okay Also light travels in a straight line except when it is reflected Or it moves from one medium to another and for this we will use Snell's laws in a second or after the break probably So it moves in a straight line then there's glass so it bends Towards normal in this case. We will see why later and then afterwards it travels parallel to the original line and again in a straight line notice that this is really a defined beam through the glass and Then afterwards is also a confined beam of light one ray of light Well, it's a beam, but let's consider it as a ray for one for practical purposes and the bending of light relates to The medium traveling with different speeds inside the medium so the light traveling with different speeds inside the medium And I'll get back to that in in the discussion part in a few slides So it bends And then the opposite effect takes place when it leaves the glass again it bends backwards well away from the normal again to have a Well propagation direction that is parallel to this one.

Yeah, this is what you've seen in secondary school as well, right? Okay, some examples of refractive indices And this depends on color as well So let's assume that we have Light of a certain color so yellow light with this wavelength 589 nanometers then Different media have different indices of refraction for this wavelength So water 1.3 glass typically Yeah around one and a half Quartz diamonds well diamond has a higher Refractive index so light slows down even more in time.

This is something you would look up in tables What we do in our well on our exams is we give if we have two media Then we specify this one here as an n_1 if it's air if it's air then it's one For air and this one typically has an n_2 and we have to work with that Don't concern too much about the numerical value You should know that one is larger than the other to know in one direction light bends, but that's about it So let's do a bit of discussion Think about glass typically and is one and a half Why is glass then transparent? Someone have an idea for that any frequencies Okay.

Yeah first. I'm right indeed glass is transparent because it does not absorb light of visible frequencies So the the band gap energy and that relates back to the semiconductor physics part the band gap energy between the energy levels of electrons in the in the Well in the shells around the atom the nucleus. Sorry It's such that visible light cannot excite These atoms into a higher state so visible light is not absorbed by glass Because the band gap energy is too high and Glass is not transparent for higher frequency light so light we don't see anymore, but if you think about x-rays or Gamma rays then glass is no longer transparent. It's just for visible light it happens that glass is Why such a combination of materials? So

it's silicon oxide basically That has a band gap energy that it such that it does not absorb Visible light.

Yeah, go ahead. Yeah. Yeah, so it is it also has to do with the geometry of the lattice So their silicon oxide in glass is bound in such a way that the band gap energy of glass Does not permit absorption of light?

Yes sand is not transparent in general. No, it's brown ish Because it has a different molecular structure.

Yes So this this depends largely on molecular structure and it's really well Physics studies being done on well on characterizing such materials for a range of frequencies And especially what what ASML has to be doing for making their tips Operate at these really really high frequencies. They have to work with materials that have the right properties at these frequencies So they need good characterization of these materials. Otherwise this Well, you absorb light that you use for a foot for the lithography. That's not what you want Okay, next question So why does light travel slower through glass? There's many Explanations for this also on on your job on the internet AI I'm not sure if they does it correctly already Why does it travel slower? Because the glass is a denser medium.

Yeah, but what does that mean? That's it's answered an air. Yes Can you specify more or not?

Yeah, go I'll proceed. Yeah, go ahead Yeah, but why would it slow down then not completely but someone else perhaps There's all kinds of explanations.

Let me use this one. There's the so-called ping-pong hypothesis So if light travels through air goes like this, but it has to if it travels through a medium Then it can bounce off Atoms in the medium and that makes the path longer so it travels slower This is not correct Because why would the light come out in this direction and it could also go off in this direction or in that direction That's not what we see if we look if we shine a beam of light through a medium that the picture I had earlier This does not look like it scatters in all directions, right? There's also the idea that light could be absorbed by an atom and that takes a bit of time and then it's Reemitted again to the next one takes a bit of time and that could also make it slow down through the material That's also not correct Because why would the light then propagate in this direction if it's absorbed it can be Re-radiated in all directions so again, then this picture would not Happen you would find a Blob of green or sorry blob of yellow light come out from here because it would scatter in all directions Yeah action Yeah, okay, then you get to Feynman integrals and things like that Perhaps at the end of today's lecture you you can relate to that concept. I don't use the word action, but it's here It's different well light does interact with the medium But not in a way that it gets reflected or bounces off We really need the wave nature of light to understand this Right is an electromagnetic wave So when it goes through glass this electromagnetic waves will wiggle all of the electrons that are connected to the atoms in the material so these atoms go around in well Circle the nucleus But they do get excited if an electric field gets past them Light is in electromagnetic waves, so it carries in magnetic and electric fields So light will wiggle all of these atoms and all of these atoms together all of these electrons together will generate counteracting electric field That has the net effect that it slows down light just a little bit inside a medium, so it interacts with The medium in a subtle way. It's not absorption and re-emission definitely not what it does interact with electric with the electrons and These oscillations for the lattice of the medium that you're looking at Create a net effect that it is effectively slowed down, so there's a Two nice videos on this by 60 symbols. I'm not sure if you know that YouTube channel Three blue one brown you probably know right.

This is a really really nice video. It's half an hour long So I'm not showing that here He really shows with his animations. I wish I could do these He shows why the wavelength is different why? the result of the oscillations of the electrons in the lattice create Well the net affected light has a shorter wavelength what you see here and slows down in glass So is that comes because of the electromagnetic interactions? And the reason why light bends at an interface becomes an interface It will go off in a different direction That also has to be described with the wave behavior of light with the electric field that depends on Well that scales with the permittivity so a material property of the medium. That's probably what you meant as well Yeah, and that bending There's also wrong explanations for that this marching shoulders analogy in which they march through Well air and through mud in mud. They are slower that would cause them to rotate. That's also not complete in explaining this effect So the 60s let me see Well there's other videos in that I think I'll link these on the on the converse page. There's two nice videos of Fermilab explaining that as well for what we need to know is we need to know that light slows down the medium and That it bends if it goes into a medium with higher refractive index and that's because of the scaling of the electric field Which relates to the boundary conditions over the interface for the electric field? I think that's a nice moment for a break. Let's continue at half past half past two Yes, we will continue I would like to continue then we can go to the midterm test well in time So let's look at how to do calculations on lights.

Let's look at how We do we work with reflection and refraction For this we need Snell's laws So let's first use some definitions we have two media Medium one and two or a and B typically a and B and There's an incident ray on here. There's a reflected ray and there's a refracted ray Each have their own angle so there's a θ_a the incident ray θ_a are reflected ray and θ_b The refracted ray and we assume that we have light of a single color so that we Well don't have to dig into account that there's multiple colors each having a different refractive index This works as a single color monochromatic light. We typically call it monochromatic single color Then we have Snell's law Snell's laws actually The first one is that all of these all of this happens in a plane This is a 2d plane in which all of this happens, so we can't have this This coming out at an angle so forward for instance it has to happen in a single plane Locally this plane can be well has to be at least perpendicular To The surface, but everything happens in that plane So we a 2d picture Suffices to do calculations on this a 2d drawing We have the angle of incidence equals the angle of reflection So that I mentioned that earlier angle of incidence equals angle of reflection. This angle is the same as this angle Because we don't change medium. So the angle is the same Always measured with respect to the normal and the normal is this line Perpendicular to the interface and then for refraction we have the law with the signs and $n_a \sin \theta_a = n_b \sin \theta_b$ so the refractive index in medium one times the sine of the angle of incidence equals The refractive index of medium B times the sine of the angle in medium B And I'll give some examples now how to work with that So this was discovered by a Dutch scientist I Can't take credit for this, but I'm still a bit proud Quite a long time back, but even earlier there was documentation of this well 600 years earlier there was documentation of this in the by a Persian scientist But Snell formalized this into the three laws Well these you probably know try to write them down by yourselves This should take half a minute only and then we proceed with the examples So let's see if you understand the concepts.

Let's do a quick quiz question So what can you say about the angle of refraction at an interface between two media? three options here you can think of which of the three laws of Snell you would need for this who thinks a B C No one dares to raise a hand for see if we got so many so much response indeed it depends on Well the refractive index in between the two media, so let's see the three typical cases that we have So in case one it goes to a medium that has a higher refractive index This can be air and then you go from air to glass which has one and a half as a refractive index And if you go to a higher refractive index you bend towards the

normal So angle θ_B is smaller than θ_A this follows from Snell's law So we had $n_A \sin \theta_A$ and $n_B \sin \theta_B$ equals $n_B \sin \theta_B$.

So it's not $\sin \theta_B$ equals n_A over $n_B \sin \theta_A$ So if n_B is higher this fraction is less than one So the sine of an angle becomes of this angle becomes smaller than the sine of that angle so the angle has to become smaller So higher refractive index you bend towards the normal If you go to a lower Refractive index you bend away from the normal that's the inverse of this well the same still holds because it has to hold Snell's law But n_B is now smaller than n_A So this fraction becomes more than one so $\sin \theta_B$ is larger than $\sin \theta_A$ meaning that θ_B has to be larger than θ_A The reflected angle does not change that's that this should also be θ_A Similarly in the previous case that is θ_A the reflected angle Also here is the same in case three then The easiest one if you have normal incidence Then you don't bend The angles are the same so it's 90 degree on the way in 90 degree in the way out Always with respect to the normal and the angle here is also 90 and that also relates to the sign. Yeah, go ahead It should be the same on both sides so We will do symbolic calculations anyway, so Shouldn't matter too much you do whatever you're used to if you think in degrees more easily then Well, we can convert degrees to radians, so that's that's fine Depending on how the angle is given Sometimes it's given in degrees, and then you can work with that in degrees Doesn't matter too much We will do symbolic calculations anyway, and especially for these problems You will need to do construction and use some trigonometry to solve the problem You will see that on Monday in the studio classroom, so this gives Effects like this right if you have something that's partly submerged in the water. It seems bent Why does that happen well it happens as follows if you look at From this point on the ruler Going out it goes from a medium with a higher refractive index to a medium of lower effective index So locally with respect to normal this angle is smaller than the angle here So it bends away from the normal so it arrives Well, it seems it arrives at your eye this point arrives Array coming from this point arrives there at your eye But for your eye it seems to have come from here Because your eye cannot process that well the law of science here So it seems that it has propagated from here So it seems that from where it's submerged in the water it has been bent upwards Because it is perceived as coming from a different point The eye thinks in straight lines, so it seems that it is observed here whereas in fact it is here And this is what birds who hunt for fish have to account for right if it's standing on the water side and wants to Stab a fish on its beak then it really has to take this into account otherwise you miss it Hunters hunting for fish have to do the same Have a spear With a shotgun and gets different Okay, let's see if you understand the concept and then I'll show a nice video in this section and a smaller and be We think the answer a answer B answer C As a D Answer E But we have we have all options so none of the above is It's not true. It is indeed answer B So we go towards a medium with a higher refractive index, so we bend towards the normal and The medium has a higher refractive index so light has to slow down While going into that mean so answer B is correct.

Let me show you the Cup and penny experiments Which I think this one just Just no real sound doesn't add a lot today in the videos Let me put it on loop, so you don't see the penny when it's lying here. Well just the rim and when you Pour water into it You get the effects exactly as I showed in the in the figure You can look more down into the cup because of water because the light bends away or the light coming from the penny bends towards you So you make use of Refraction to be able to see this Right my kids like that when I show them this experiment So let's go to another principle to explain basically Snell's laws, and that's called Huygens principle Huygens's principle philosophy s So another way to look at this to construct solutions to such problems is to consider every point on a wavefront So bear in mind wavefronts is in equity phase plane of a wave propagating so well say all the Local maxima the troughs of a wave propagating rays are perpendicular to that You can consider all the points on the wavefront wavefront as a wavelet that

spreads out light in all directions and The speed is equal to the propagation of the wave propagates the speed of the wave in that medium So in air at the speed of light V is C But in general in the medium, that's less than C so all of these points will generate such a circular wavelets going outwards and By taking down a snapshot at a new time instance You can calculate a well determine the locus of all of these points again to see where we are then And what effectively what has happened is that you have propagated in the direction of the ray That's it. That's kind of a step in between to construct such wavefronts if you think about a wavefront striking a surface Then For reflection you have to take into account that it can change direction so It is incident on the surface so in this and if you construct Well for every wavefront such a wavelet that travels VT over time T Then well at this point for instance in this point these have Touched the surface again. So a wavelet from this point will generate a curve well Going out from this point, right? So let's construct that so a if you look at this wavefront At this time and sent it as a source point that we use exactly on the interface And to do calculations, then we propagate this one outward also in all directions. Well not towards The object because it reflects but outwards upward in this case and this will then create a well at V times T It did will travel this distance This point has also traveled in at same time V times T .

So This point V times C gives well this semi circular arc This point in V times C gives this semi circular arc and these have to be on the same line because they belong to the same New wavefront so we can draw a line connecting this point and that point So there's many solutions, of course, and that's also one disadvantage of the Huygens principle If you do this with with enough sources, you can construct a nice new way And in this way you will find that the reflected ray will be at a different angle than the incident ray So let's do this construction in detail Let's try to look at how this happens So we are going to zoom in on this red Triangle that was connected to the wavefront going out from here and away from coming in well from from from here Going in basically Let's construct that So we zoom in on that triangle now And we see that well in a time T .

It has traveled this distance OB And this wavefront has traveled the same distance from A to Q also VT . So that's also the two Circular well circular arcs that we can draw this one coming in from here. So there's a circular arc here And this one coming out from here. So there's a circular arc here Well, and these angles these triangles have to be well similar triangles so triangle This one so let's let me do the correct order.

So AOP so it's on top luckily so AOP has to have a similar shape as OAQ So these two triangles have to be of similar shape They have a right angle here because this is how the ray propagates perpendicular to the wavefront same here This is perpendicular to the new wavefront so perpendicular to the wavefront. So these triangles have to have the same well form which means that all of their angles have to be the same well on this Acute angle in this triangle in triangle a OAQ has to be the same as the angle here in AOP And that gives us that the angle of incidence has to equal the angle of reflection So this is also While confirming that the angle of incidence is the angle of reflection, but now using Huygens forces So bear in mind the construction for this was that we Looked at incident rays reflected rays and we used Points on this wavefront as a Huygens force creating a wavelet propagating outward with a velocity V We're in the same medium. So the velocity is the same and over the same time we travel the same distance or the same radius And that way we can find One of Snell's laws It's okay.

If you can follow the reasoning here, that's sufficient for for this lecture. We will use Snell's laws. We Do not need Huygens forces to do calculations We will use Huygens forces a bit more when we talk about interference and diffraction But that's something I'll cover on Monday This concept clear not completely but Yeah, please look in the book as well that book may Sometimes have a different

explanation under I give here. So it has the advantage that you get Two different explanations one of which may resonate better with your your mind.

We can also look at waves Striking an interface so we have two materials now we have this situation medium a medium B And in medium a there's a higher velocity So typically air medium B is let's call it class the velocity is lower so within the same time in medium a we travel more distance than a medium B and Also here we can do a detailed construction by looking at Well, basically this part here We're assuming in on this part And look at what happens there so This distance is different from distance because the speeds in the media are different, right? Still we have triangles that look kind of similar because they share this hypotenuse OA is the same for both triangles.

These are right angles So we can define the sine of theta B as being AB over OA That's the sine of theta B. The sine of theta A is A prime O over OA.

So that's VAT over OA And then we can equate OA from this equation to find Nelsbar Sorry, yeah, you can imagine that you get to this if you equate OA Where you take into account that the velocity of the medium Scales with the refractive index.

So VA That's the ingredient you need VA C over N A VB equals C over N B And together with these two You can create Snell's law because T drops out C drops out and OA You will eliminate from the equation and then you get Snell's law But now from a Huygens point of view, is that clear? The important part is that you can work with Snell's laws This is another way of looking at them, but Huygens force is we need later also for a double slit experiment And that's why I need them to at least be introduced here Okay, please try to summarize this as well. What is a Huygens force? How does it work? Why is it useful? We'll go to some examples.

Well a nice example of something you can explain with Huygens principle is A mirage, so if You're in a really hot desert you can see Phantom reflections of things on the soil And this works as follows Well, I say examples of Huygens principle can be explained using Huygens principle, let's face it like that What you see is that a Wayfront higher up I'll say the head of the camel or the tip of a mountain you see that directly it travels straight on through air and that's higher up It can also travel Because there's a gradient in the air so that has cooler air Higher up hotter air near the ground because the desert sand is really really hot So there's a change of refractive index a continuous change of refractive index over the height above the ground And that makes that well depending on Where the ray comes from it may have traveled well from a different point of view Well, that's why you can see a reflection of the mountains here also Well as coming from the sand so it is this has the the illusion of water and lying there So we can mistake that for a sheet of water that is lying there that comes from Well a gradient in the refractive index over the air There's a nice video on that With a similar type of Thing so let's see we've seen this one Let's do this one first.

There's light bending so it's a laser. So it should send out a straight Sorry for this should send out a straight beam But you see that it bends so it's not just straight it curves and that is caused by the fact that there is a Gradient in the sugar in the water and similar here, so this is a Straight line, but it's not always in a straight line.

It can even seem to bounce And this is god, but of course by similar effect as that we had with the the mirage There's a gradient in the sugar content But from future on top, it's a straight line. It's just a gradient in vertical direction This is weird, right?

It is caused by a gradient in the water causing a gradient in refractive index and that makes that you kind of create a waveguide based on the The medium properties

I'm really happy that the sound in this video has an annoying music you see that it bends already So that's the same effect as the mirage we had earlier Careful wake up.

So let's see if you understand the concepts This is kind of also an analogy that is often used to why light bends It does not explain How much it bends it does explain some aspects of it. Let's see if we understand this one You can have sprinters on a parking lot. You can have a swimmer In the sea and a lifeguard on the beach who has to run to the swimmer and then swimmer also a certain distance Swimming is slower than running. So you have the same effect There's a medium on which you have a high velocity and a medium in which you have a low velocity So is it then worthwhile? Running shortest distance than running over sand Or running longest distance over the parking lot and then just the shortest path to the sand who thinks path A is quickest path B See D E Okay Intuition of most of you is right Or F. Sorry No, they don't take all they don't all take the same time. That's obvious Somewhere in between C and E there should be the optimal well the the fastest path and This is exactly what light also does if there's medium of different refractive index indices Takes the shortest time to get from P to Q How does light know that?

Well, that's it's a law of nature. It does this Automatically it happens like that.

It can be explained using this Using shortest time principle.

So that probably relates to the least action explanation as well But it does not explain the angles you could do a calculation on this actually might be a nice new example Well, we'll see but Light does this as well takes the shortest path Another one Well, we're getting near the end. That's good Let's see if you recall what I mentioned earlier.

Where do you see the fish who thinks answer a answer B? also see indeed the construction was like this a Ray of light coming from the fish goes from high refractive in this index lower refractive index. So it bends away from the normal So it seems that is if it has come from well the eye that things in straight lines So from somewhere up here, okay If you look at a sunset That's safe to do because a lot of the harmful radiation is gone already.

Yeah question Yeah Yeah, good point Probably just vertically upward. It should be in the in the plane of you and the fish and I only vertically upward but if the water is Choppy so with waves then it's really difficult to do this.

So probably I Would think just straight up but whether you see that yeah that then it also seems smaller So you misjudge in different ways how you see the fish? It might be smaller than you think if you think it's it's bigger than as you see it Then you may mistake it for further away Closer by I mean so that yeah, there's a trade-off also be between the interpretation of the viewer and where the fish is Let me get to the sunset So this seems flattened from the bottom side if you try to draw a circle It seems flattened somehow and that is also caused by refraction And the reason is that the Sun Well say Sun is Well comes from far away. It has a size a certain size an array from the top of the Sun Comes at a different angle through the atmosphere and an array from the bottom of the Sun so Depending on where the ray enters the atmosphere and this is why it's apparent at sunset then You have the effect of the curvature of the atmosphere more if it's straight up you well the Sun is straight up And all of the Sun beams come Perpendicular down to you as an observer, but if it's at an angle so at the sunset then it seems as if lower parts of the Sun Go through a different layer of atmosphere Also at a different angle and this causes a varying normal to the interface so a various varying index A varying angle of refraction depending on the ray where it comes from so ray from the top have seen Well come from higher up so they come from Higher up than layers than rays from the bottom they come from here for instance, and they also arrive at your

eye But at a different angle, so that's why you see that the Sun flattens at sunset, okay? Why this is red I'll come back here to next lecture So the final topic then is total internal reflection, so and then I'll manage before yeah way before quarter past So remember there is one case in which light bends away from the normal that is if you go from a high refractive index to a lower refractive index if you do this at a Too large angle so if one perpendicular incidence it continues perpendicularly If you increase the angle then there is a certain angle where it bends so much does it that it doesn't leave medium a anymore? So it stays in medium a does not go into medium B So this is at the so-called critical angle So when θ_B equals 90 degrees so when the sine θ_B is One so then we have an a sine θ_a equals and B From that we can calculate the critical angle, so θ_a the critical angle This is the incident ray is then and B over and a So this happens well this can only happen if And B is lower than and a because the sign of something always has to lie between minus one and one So this has to be cannot be larger than one then you don't find a solution, so we have to go from Lower to higher refractive index then you get a ratio.

That's less than one For every angle that is larger than this critical angle, then you get total reflection So in that sense that kind of answers your question as well When do you have total reflection that that happens in this case? But in general how much has been reflected how much has been refracted? That's something that you really learn in the electromagnetic scores not in this course, so how does that work? Well, let's use an animation that tries to build it up if the ray perpendicular You get reflection and refraction, but it is at right angle, so you don't see that If you increase the angle you get some something refracted Something reflected if you increase the angle You end up at an angle where it doesn't enter medium 2 anymore So we have At this case you have one ray of light Propagating along the interface, and then you still have the reflection and if you go further you only have The reflection they have total internal reflection and sometimes we use this in problems as well because that's something you can You can do calculations with for what angle do I have to shine light through a prism to get? Total internal reflection at the far end of the prism for instance so some nice example here There's light shining on three Mirrors tilted at a different angle well for one of the Well at the critical angle there's light also propagating along the interface, but it's that's probably this one But you still have the reflection also for these you have reflections, so that's the reflection from this one This gives a reflection there and this gives Total internal reflection, so this is the reflection. This is the The ray of light that is not propagating into air So just it stays in the water.

I think this was also covered in the secondary school At least for those who had optics there Total internal Total internal reflection is used a lot if you think about optic fibers That's you want to confine light in the fibers you want to transmit it as far as possible Without it leaving the fiber so you make use of total internal reflection there's also If you have a beam of water pouring out of an aquarium for instance You can have the laser being totally in for internally reflected in the Sorry, let me play that again In the beam of water.

I think I have another one so it doesn't leave the water it Due to total internal reflection it stays inside the water And that's how an optical fiber works as well There's another one Again annoying sound luckily.

I switched it off here. You see a green laser.

That's confined in the In the water that's pouring out of a Well a bottle in this case you have to hold your hand Perfectly still otherwise you get out of the beam That's right, okay This concept is used also in a lot of applications if you have a reflector on the back of your bike then well You make use of that concept that all the light coming in from whatever angle is reflected So it lights up if you shine

light on it. That's exactly the purpose. Impurities this can also be used. This is also what makes diamonds really shine. Diamonds are special in that sense because they have a high refractive index. So it's like glass typically has one and a half diamonds. Well 2.4 typically. So there is a very small critical angle. It reflects light really easily from whatever angle it comes in and then the way it's got even emphasizes that more so and well it should maximize light return through the top. That's why they sparkle. Okay, and actually I should have shown the movies here. This is what happens in fibers. You want to trap the light in the fiber. So it should always be a higher density. Sorry higher refractive index core compared to the cladding around the core. Because you have to have total internal reflection happens if you go from a high refractive index to a lower refractive index. That's how this works.

I think you want to go to a test, right? so to remember we looked at the nature of light reflection and refraction. Derived in two ways now. Snell's law and Huygens principle and we looked at total internal reflection and next Monday. We'll continue this with dispersion and also interference. Good luck with the midterm test. I hope you all are here to make that. To do the test please go to the room assigned to you and then well, I'll see you on Monday. Good luck.